UBP Client Attrition API

Rule-Based Product Recommender System

Technical & Business Documentation

1. Overview

The recommender system is a rule-based engine embedded in the UBP Attrition API. It runs immediately after the attrition prediction pipeline and produces a prioritized list of up to five UnionBank product recommendations for each client. Its purpose is to give relationship managers a concrete, actionable next step based on the client's financial profile and predicted attrition risk.

1.1 Why Rule-Based and Not ML-Based?

The EDA phase of the attrition model project established that all available features carry near-zero discriminative signal for churn prediction (Cohen's d < 0.03 for all continuous features; Cramér's V < 0.01 for all categorical features). The best model achieved a PR-AUC of 0.0495, which is approximately equal to the random baseline of 0.05 on a 5% positive-rate dataset.

Building a second ML model on top of an already weak signal would amplify noise rather than add value. Rule-based logic anchored on observable, auditable client attributes (i.e. income, card tier, tenure, engagement) is more interpretable, more sensible to segment heads, and more immediately actionable for the campaigns team.

2. System Architecture

The recommender is entirely downstream of the attrition pipeline. It does not affect model training, feature engineering, or serialization. The data flow is:

```
Raw client features (11 fields)
        │
        ▼
  pipeline.predict_proba()        ← sklearn inference pipeline
        │
        ▼
  attrition_probability (float)
        │
        ├── + original raw features
        ▼
  get_recommendations()           ← api/recommender.py
        │
        ▼
  RecommendationBundle            ← returned in API response
```

The recommender receives the attrition probability and all 11 raw input features. It derives additional signals internally (income tier, risk tier, life stage, credit utilization) and applies five independent rule layers.

3. Input Features Used by the Recommender

The following features are passed from the API to the recommender. Features marked 'Unused' are accepted for forward compatibility but intentionally excluded from rules. The EDA process confirmed Cramér's V < 0.01 for Gender, MaritalStatus, EducationLevel, and Country, meaning they have no meaningful relationship with any outcome.

Feature	Type	Used In Rules	Reason If Unused
attrition_probability	float [0,1]	Yes	
Age	int	Yes	
Income	float (PHP/yr)	Yes	
CreditLimit	float	Yes	
TotalTransactions	int	Yes	
TotalSpend	float	Yes	
Tenure	int (months)	Yes	
CardType	str	Yes	
Gender	str	No	Cramér's V < 0.01 in EDA
EducationLevel	str	No	Cramér's V < 0.01 in EDA
MaritalStatus	str	No	Cramér's V < 0.01 in EDA
Country	str	No	Target-encoded SD = 0.0068

4. Derived Signals

Before the rule layers execute, four signals are derived from the raw inputs. These are computed at runtime — not features from the ML pipeline.

Derived Signal	Formula / Logic	Purpose
income_tier	Entry (<360k) / Standard (360k–1.2M) / Premium (1.2M–3M) / Wealth (>3M)	Primary driver for product eligibility and cross-sell targeting
risk_tier	Critical (≥0.65) / High (≥0.45) / Moderate (≥0.25) / Low (<0.25)	Overlay for retention rule activation. Acknowledged as weak signal.
utilization	TotalSpend / CreditLimit	Engagement health indicator. < 15% = underusing; > 85% = credit stress.
life_stage	Age < 30: Young Professional; 30–39: Early Career; 40–54: Mid Career; 55+: Pre-Retirement	Drives life-stage product recommendations (EON, home loan, BancAssurance)

5. Income Tier Thresholds

Thresholds are derived from Bangko Sentral ng Pilipinas (BSP) consumer banking income segmentation guidelines and UnionBank's published card eligibility requirements. The dataset Income field is treated as annual PHP income.

Tier	Annual Income (PHP)	Monthly Equivalent	Typical Products
Entry	< 360,000	< 30,000	EON, Blue Card, basic savings
Standard	360,000 – 1,200,000	30k – 100k	Silver Card, personal loan, auto loan
Premium	1.2M – 3,000,000	100k – 250k	Gold Card, UITF, home loan, Priority Banking
Wealth	> 3,000,000	> 250k	Black Card, Wealth Management, GlobalLinker

6. Card Upgrade Eligibility

The recommender follows UnionBank's published card hierarchy: Blue > Silver > Gold > Black. Upgrade eligibility is checked by comparing the client's annual income against the minimum income requirement for the next tier. Only one upgrade step is recommended at a time.

Upgrade To	Minimum Annual Income (PHP)	Monthly Equivalent
Silver	480,000	PHP 40,000
Gold	1,200,000	PHP 100,000
Black	3,000,000	PHP 250,000

7. Rule Layers

Five independent rule layers are evaluated for every client. Results from all layers are pooled, deduplicated, sorted by priority, and capped at five recommendations.

7.1 Retention Rules (_retention_rules)

Triggered by elevated attrition probability. Given the model's near-random performance, these are intentionally conservative — only activating at probability ≥ 0.45 to avoid flooding every client with retention offers.

Condition	Product Recommended	Priority
probability ≥ 0.65	UnionBank Rewards Points Multiplier	Critical
probability ≥ 0.45 AND tenure ≥ 36 months	UnionBank Loyalty Cashback Program	High
probability ≥ 0.45 AND income_tier = Premium/Wealth	UnionBank Priority Banking	Critical

Note: Retention rules are the weakest layer by design. The attrition probability is near-random (PR-AUC ≈ 0.05). These recommendations function right now serves as POC, but ca be utilized once better data arrives.

7.2 Card Upgrade Rules (_card_upgrade_rules)

Objective, income-gated eligibility check. This is the most reliable rule layer because it is based on verifiable financial data with no dependency on the weak attrition signal.

Condition	Product Recommended	Priority
Current card = Blue AND income ≥ 480k	UnionBank Silver Visa Card	Medium
Current card = Silver AND income ≥ 1.2M	UnionBank Gold Visa Card	Medium
Current card = Gold AND income ≥ 3M	UnionBank Black Visa Card	Medium

7.3 Engagement Rules (_engagement_rules)

Behavioral engagement signals from transaction frequency and credit utilization. These are the most actionable rules because they are directly observable and map to specific product interventions.

Condition	Signal Meaning	Product Recommended	Priority
Credit_Utilization < 15%	Card underused	UnionBank PayDay Promo / Merchant Cashback	Medium
TotalTransactions < 24	Digitally disengaged	UnionBank Online / EON Digital Banking	Medium
Credit_Utilization > 85%	Near credit limit	UnionBank Credit Limit Increase	High

7.4 Cross-Sell Rules (_crosssell_rules)

Proactive product matching based on income tier, age, and tenure. Designed to deepen the client relationship and increase switching cost — both reduce attrition risk without relying on the weak ML signal.

Condition	Product Recommended	Priority
income_tier = Standard/Premium/Wealth AND tenure ≥ 12mo	UnionBank Personal Loan	Low
Age 28–50 AND income_tier = Premium/Wealth	UnionBank Home Loan	Medium
Age 25–45 AND income_tier = Standard/Premium/Wealth	UnionBank Auto Loan	Low
income_tier = Wealth OR (Premium AND utilization < 40%)	UnionBank UITF	Medium
Age ≥ 35 AND income > 1.5M AND income_tier = Premium/Wealth	UnionBank Business / GlobalLinker	Low

7.5 Life Stage Rules (_lifestage_rules)

Products aligned to the client's life stage. These are triggered independently of attrition probability.

Condition	Life Stage	Product Recommended	Priority
Age < 30 AND income_tier = Entry	Young Professional	UnionBank EON Cyber Account	Low
Age ≥ 55 AND income_tier = Standard/Premium/Wealth	Pre-Retirement	UnionBank Time Deposit / BancAssurance	Medium

8. Output Structure

Each recommendation contains five fields designed for direct use by a relationship manager or downstream CRM system.

Field	Type	Description
product	str	Full UnionBank product name
category	str	Retention | Upgrade | Engagement | CrossSell | LifeStage
reason	str	Plain-language explanation for the RM -> references the specific signal that triggered the rule
priority	str	Critical | High | Medium | Low -> determines sort order
action	str	Specific next step for the relationship manager or system to execute

Recommendations are deduplicated (same product cannot appear twice), sorted by priority (Critical first), and capped at 5 per client.

9. Priority Ordering & Deduplication

After all rule layers are evaluated, results are merged into a single list. The following process is applied before returning:

- Sort by priority: Critical (0) > High (1) > Medium (2) > Low (3)
- Deduplicate: if the same product name appears from multiple rules, only the first (highest priority) instance is kept
- Cap at 5: only the top 5 unique recommendations are returned per client

This ensures relationship managers receive a focused, prioritised shortlist rather than an overwhelming list of every eligible product.

10. Sample API Response (Single Client)

Input: 47-year-old male, PHP 1.6M income, Gold card, 3 transactions, 46 months tenure.


```
{
  "attrition_probability": 0.0521,
  "attrition_flag": 0,
  "attrition_risk_tier": "Low",
  "client_segment": "Mid Career · Premium",
  "top_drivers": {
    "Avg_Txn_per_Tenure_Year": -0.1832,
    "TotalTransactions": -0.1204,
    "Age": 0.0871,
    "Income": -0.0634,
```

```
    "Credit_Utilization": -0.0512
  },
  "recommended_products": [
    {
      "product": "UnionBank Black Visa Card",
      "category": "Upgrade",
      "reason": "Income qualifies for Black tier (PHP 1,600,000/yr)",
      "priority": "Medium",
      "action": "Initiate card upgrade offer to UnionBank Black Visa."
    },
    {
      "product": "UnionBank Online / EON Digital Banking",
      "category": "Engagement",
      "reason": "Only 3 total transactions recorded — less than 1 per
month",
      "priority": "Medium",
      "action": "Send digital onboarding push notification."
    },
    {
      "product": "UnionBank Unit Investment Trust Fund (UITF)",
      "category": "CrossSell",
      "reason": "High-income client with moderate card utilization",
      "priority": "Medium",
      "action": "Refer to Trust & Investment Division."
    }
  ]
}
```


11. Limitations so far and future improvements

11.1 Current Limitations

- Income thresholds are fixed constants > not dynamically calibrated to portfolio-level data
- Attrition probability is near-random (PR-AUC ≈ 0.05); retention recommendations are soft signals only
- No personalization beyond the available features. Product affinity scoring not possible without transaction history
- Card eligibility requirements are approximate. Subject for approval from the stakeholders.

11.2 Recommended Enhancements

- Integrate transaction-level behavioral features (recency, frequency, monetary) to improve both attrition signal and recommendation relevance
- Replace static income thresholds with dynamic percentile-based segmentation recalculated at each model refresh cycle
- Add a product exclusion list. Do not recommend products the client already holds

12. Code Reference

File	Purpose
api/recommender.py	All rule logic then runs get_recommendations(), five rule layer functions, RecommendationBundle
api/schemas.py	Pydantic models. RecommendedProduct, PredictionResponse (includes recommended_products field)
api/transformers.py	Custom sklearn transformers like WinsorizationTransformer, LRFeatureEngineeringTransformer
main.py	FastAPI app. This calls get_recommendations() after pipeline.predict_proba()
scripts/serialize_model.py	Training script. Builds and saves the ML pipeline to models/
scripts/monitor_drift.py	PSI-based drift monitoring. Compares recent predictions against training profile, saves weekly report to logs/